

Simulation-Based Inference for an Adaptive-Network SIR Model

Florian Tobias Dreyer (A0338032M) Michelle Than Yi Xuan (A0338187R)
[TODO: Group member 3 (Student ID)]

1 Introduction

[TODO: Model motivation, data description, why likelihood is intractable. Suggested length: ~1 page.]

2 Model and Data

[TODO: Formal model description: Erdős–Rényi network ($N = 200$, $p = 0.05$, 5 initially infected), three synchronous operations (Infection, Recovery, Rewiring), and why the adaptive rewiring makes the likelihood intractable. Reference Gross et al. (2006). Suggested length: ~0.5 pages.]

The three unknown parameters have independent uniform priors: $\beta \sim \mathcal{U}(0.05, 0.50)$, $\gamma \sim \mathcal{U}(0.02, 0.20)$, $\rho \sim \mathcal{U}(0.0, 0.8)$, with prior standard deviations $\sigma_\beta^{\text{prior}} = 0.130$, $\sigma_\gamma^{\text{prior}} = 0.052$, $\sigma_\rho^{\text{prior}} = 0.231$. The observed dataset comprises $R = 40$ independent replicates, each providing an infection time series, a rewiring time series, and a final degree distribution (Figure 1).

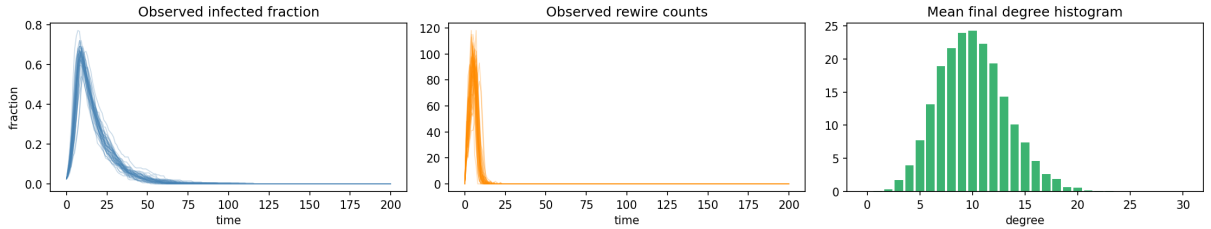


Figure 1: Observed data across 40 replicates: infection fraction (left), rewiring events (centre), final degree distribution (right).

3 Summary Statistics

Because acceptance probability in ABC falls exponentially with summary dimension (Blum et al., 2013), we compress the raw time series to a small set of scalar features. A naive approach is to concatenate all available scalars into one vector. The problem is that in a standardised Euclidean distance, a modality with more features contributes proportionally more to the total distance — a simulation can be accepted purely because its infection curve is close, even if its rewiring and degree statistics are far off. We therefore design a *balanced* set of 15 features with deliberate equal representation across the three modalities: 5 infection, 4 rewiring, and 6 degree summaries, all averaged across replicates. Within each group, features are chosen for identifiability rather than completeness (e.g. `epidemic_duration` over `final_inf` to better

separate β from ρ ; cross-replicate standard deviations to capture stochastic variability; degree quartiles and entropy over isolated/ high-degree fractions to describe the full distribution shape).

3.1 Identifiability diagnostics

Projecting the $M = 10\,000$ prior-simulation summaries to 2D via PCA and colouring by parameter value (Figure 2) reveals clear gradients for γ , moderate structure for β , and weak but non-trivial structure for ρ . This confirms the central identifiability challenge: β and ρ both suppress the epidemic — via higher transmission rate being countered by faster recovery (γ) or network rewiring (ρ) — so infection-curve summaries alone cannot fully separate them. Rewiring summaries carry independent information about both β and ρ .

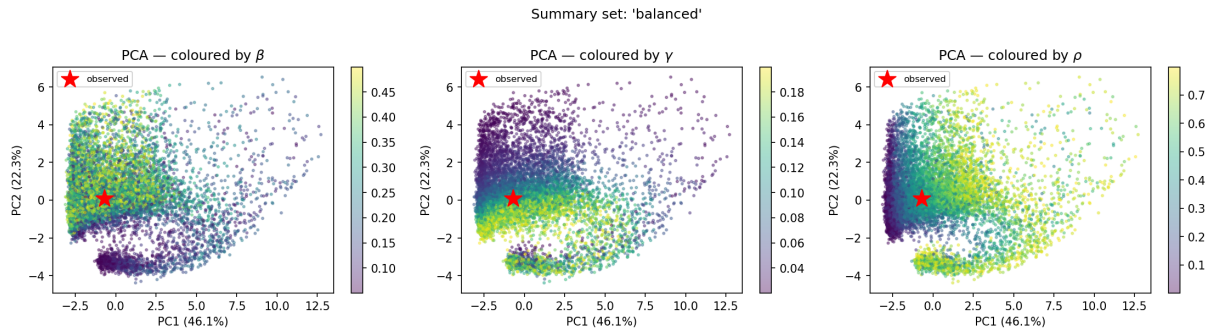


Figure 2: PCA diagnostic: prior simulations coloured by β , γ , ρ (left to right). Clear gradients for γ ; partial confounding visible in β – ρ panels.

3.2 Summary-set comparison

We evaluated the single-source sets against the balanced set at $q = 1\%$ to understand which data streams carry information about which parameters. Table 1 and Figure 3 report posterior std relative to prior std (lower = more informative).

Table 1: Relative posterior std (std / prior std) per summary set at $q = 1\%$. Bold marks the best single-source set for each parameter.

Set	β	γ	ρ
infection_only	0.377	0.084	0.854
rewiring_only	0.216	0.804	0.173
degree_only	0.924	0.932	0.483
balanced (joint)	0.479	0.378	0.262

The single-source results reveal a clear separation of information: γ is almost entirely encoded in the infection curve (*infection_only*: 0.084), while β and ρ are best identified from rewiring summaries (*rewiring_only*: 0.22 and 0.17 respectively). The degree distribution alone is largely uninformative for β and γ but provides moderate constraint on ρ (0.48).

Crucially, no single-source set identifies all three parameters simultaneously. The *balanced* set is not the best for any individual parameter in isolation, but it is the only set that constrains all three to well below the prior width. This is precisely the effect of balanced modality representation: each data stream contributes to the joint distance rather than any one dominating it.

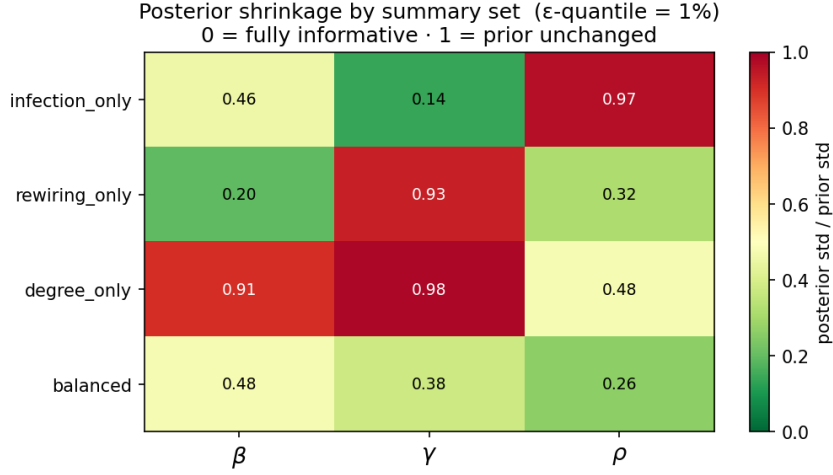


Figure 3: Posterior std (relative to prior) per summary set and parameter at $q = 1\%$. Lower is more informative.

4 Rejection ABC

We drew $M = 10\,000$ prior samples, computed the standardised Euclidean distance between each simulated and observed summary vector, and accepted draws below the 1% empirical quantile (Pritchard et al., 1999), giving $\epsilon = 1.2852$ and $n_{\text{acc}} = 100$. The 1% quantile balances a variance–bias tradeoff: tighter tolerances reduce the approximation bias (accepted draws are closer to the true posterior) but yield fewer samples and a noisier posterior estimate, while looser tolerances give more stable estimates at the cost of a prior-biased posterior. This tradeoff motivates regression adjustment (Section 5.3), which corrects for the residual bias and allows a looser 5% tolerance. Figure 4 shows the marginal posteriors; Table 2 gives the numerical summaries.

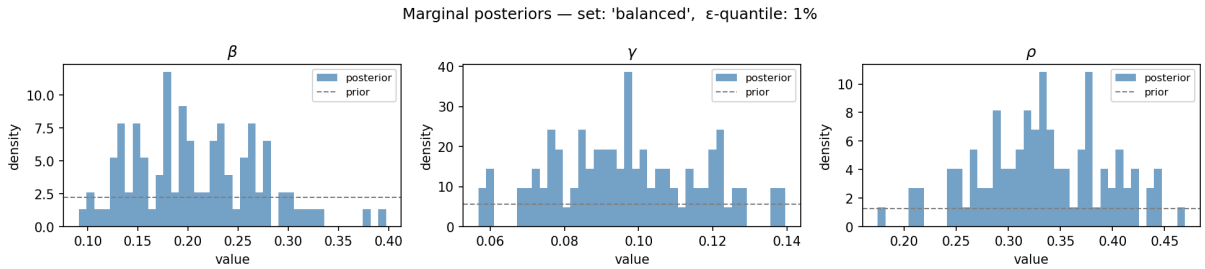


Figure 4: Rejection ABC marginal posteriors (blue) vs. uniform prior (dashed), *balanced* set, $\epsilon = 1.2852$, $n_{\text{acc}} = 100$.

Table 2: Rejection ABC posterior ($q = 1\%$, *balanced*).

Param	Mean	Std	25%	Median	75%
β	0.208	0.063	0.159	0.201	0.254
γ	0.097	0.020	0.083	0.096	0.112
ρ	0.334	0.061	0.291	0.332	0.378

γ is best-identified (directly controls epidemic duration), β moderately so, and ρ is hardest to constrain — consistent with the PCA diagnostic and the negative (β, ρ) posterior correlation driven by their shared epidemic-suppression effect.

5 Advanced Methods

5.1 Semi-Automatic ABC (Fearnhead and Prangle, 2012)

Fearnhead and Prangle (2012) show that the optimal ABC summaries are the posterior means $\mathbb{E}[\theta \mid x]$, which can be approximated by fitting Ridge regression from the balanced features to each parameter on the pilot draws. In-sample R^2 values confirm strong predictability: $R_\beta^2 = 0.812$, $R_\gamma^2 = 0.857$, $R_\rho^2 = 0.979$. Running rejection ABC in the resulting 3-dimensional projected space ($\varepsilon = 0.520$, $n_{\text{acc}} = 100$) substantially tightens the posterior for β and provides moderate gains for γ and ρ (Figure 5, Table 3).

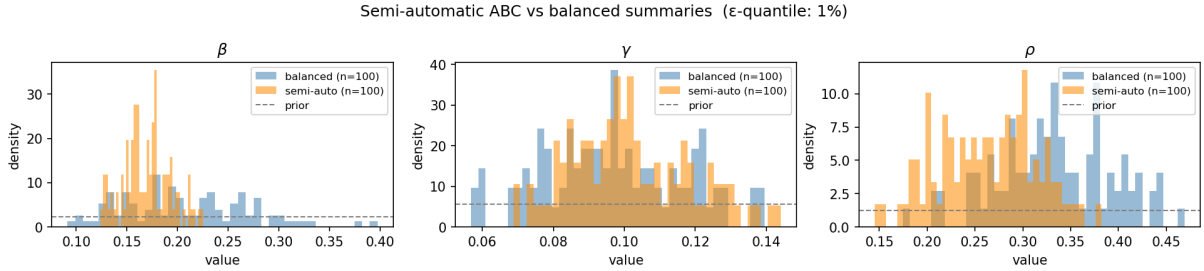


Figure 5: Balanced rejection ABC (blue) vs. semi-automatic ABC (orange), $q = 1\%$.

5.2 LASSO-Based Summary Selection (Blum et al., 2013)

Fitting LassoCV from summaries to each parameter on the pilot simulations retained 14/15 features for β ($\alpha = 6 \times 10^{-4}$) and all 15 for γ and ρ ($\alpha = 8 \times 10^{-4}$). No feature is entirely redundant, so the LASSO-selected set is effectively identical to the balanced set and yields the same posterior (Table 3).

Table 3: Posterior std and relative contraction (std/prior std) for rejection ABC variants at $q = 1\%$.

Method	β		γ		ρ	
	Std	Rel.	Std	Rel.	Std	Rel.
Prior	0.130	1.00	0.052	1.00	0.231	1.00
Balanced (15d)	0.062	0.48	0.020	0.38	0.061	0.26
LASSO (15d)	0.062	0.48	0.020	0.38	0.061	0.26
Semi-auto (3d)	0.023	0.18	0.017	0.33	0.051	0.22

5.3 Regression Adjustment (Beaumont et al., 2002)

After accepting draws at a loose tolerance ($q = 5\%$, $\varepsilon = 1.706$, $n_{\text{acc}} = 500$), each accepted $\theta^{(i)}$ is corrected for the residual gap between its simulated summary $s^{(i)}$ and s_{obs} :

$$\tilde{\theta}^{(i)} = \theta^{(i)} - \hat{B}^\top (s^{(i)} - s_{\text{obs}}),$$

where $\hat{B}$ is estimated by weighted least squares with Epanechnikov kernel weights (Beaumont et al., 2002). Figure 6 and Table 4 show that adjustment dramatically sharpens the posterior (σ_β : 0.099→0.024, σ_γ : 0.033→0.007, σ_ρ : 0.087→0.008). Crucially, the adjusted posterior is nearly tolerance-invariant for $q \lesssim 10\%$ (Figure 7), whereas the raw posterior degrades rapidly at loose tolerances.

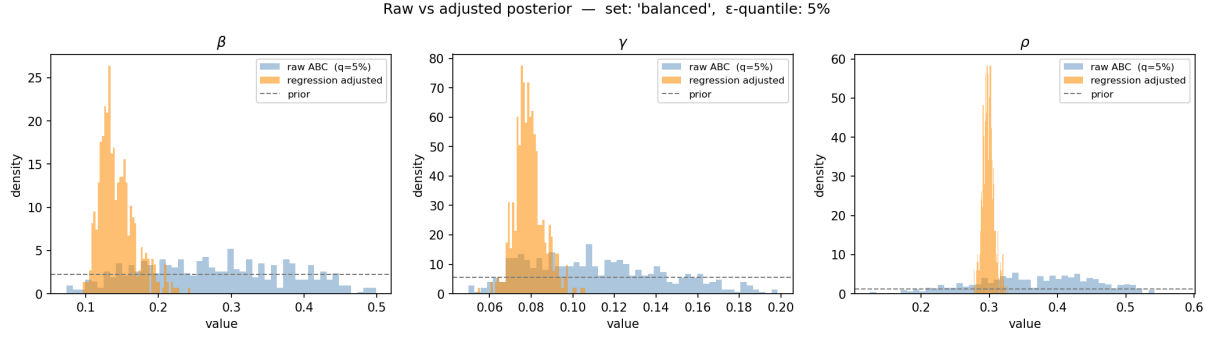


Figure 6: Raw ABC (blue) vs. regression-adjusted (orange) posteriors, $q = 5\%$, *balanced* set.

Table 4: Raw vs. regression-adjusted posteriors ($q = 5\%$).

Method	Param	Mean	Std	5%	Median	95%
Raw ($q = 5\%$)	β	0.276	0.099	0.127	0.272	0.439
	γ	0.111	0.033	0.066	0.108	0.169
	ρ	0.367	0.087	0.216	0.373	0.504
Adjusted	β	0.143	0.024	0.112	0.138	0.191
	γ	0.079	0.007	0.069	0.078	0.092
	ρ	0.299	0.008	0.286	0.298	0.313

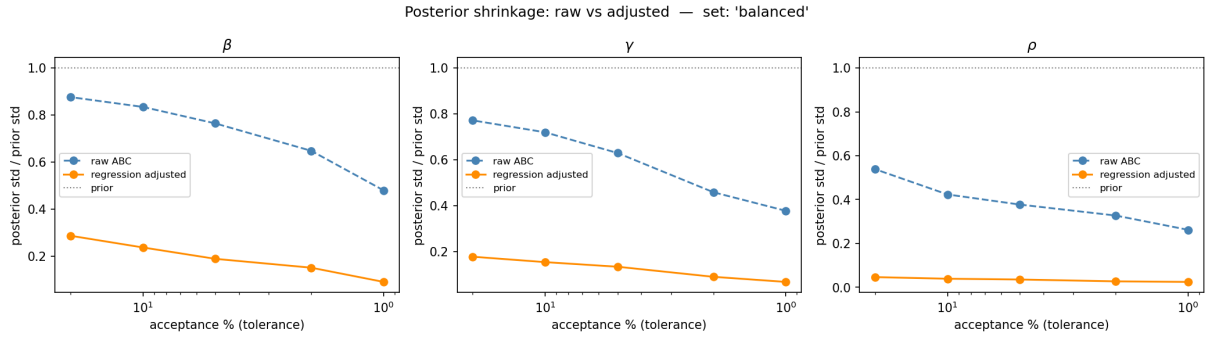


Figure 7: Posterior std (relative to prior) vs. tolerance q for raw (blue) and adjusted (orange) ABC. Adjustment is approximately tolerance-invariant for $q \lesssim 10\%$.

5.4 Synthetic Likelihood (Wood, 2010)

At each MCMC proposal θ , we simulate $N_{\text{sim}} = 150$ datasets ($R_{\text{sim}} = 40$ replicates each), giving an N_{sim}/p ratio of 10.0 for our $p = 15$ summary features, and approximate the intractable likelihood as a multivariate Gaussian fitted to the N_{sim} summary vectors, using Ledoit–Wolf shrinkage for the covariance (Ledoit and Wolf, 2004):

$$\log \hat{p}_{\text{SL}}(s_{\text{obs}} \mid \theta) = \log \mathcal{N}(s_{\text{obs}}; \hat{\mu}(\theta), \hat{\Sigma}(\theta)).$$

This is embedded in a Metropolis–Hastings MCMC with an **Adaptive Metropolis** (AM) proposal (Haario et al., 2001). Rather than fixing step sizes in advance, AM updates the proposal covariance online after an initial warm-up of $t_0 = 100$ steps:

$$\Sigma_{\text{prop}}(t) = \frac{2.38^2}{d} \text{Cov}(\theta_1, \dots, \theta_t) + \epsilon I, \quad d = 3,$$

so the chain learns the posterior shape automatically and no manual step-size tuning is required (Roberts et al., 1997). Configuration: $N_{\text{steps}} = 2000$, burn-in = 500, initialised at the regression-adjusted posterior mean (0.143, 0.079, 0.299). The chain achieved an acceptance rate of 24.0%, consistent with the optimal AM target (Roberts et al., 1997).

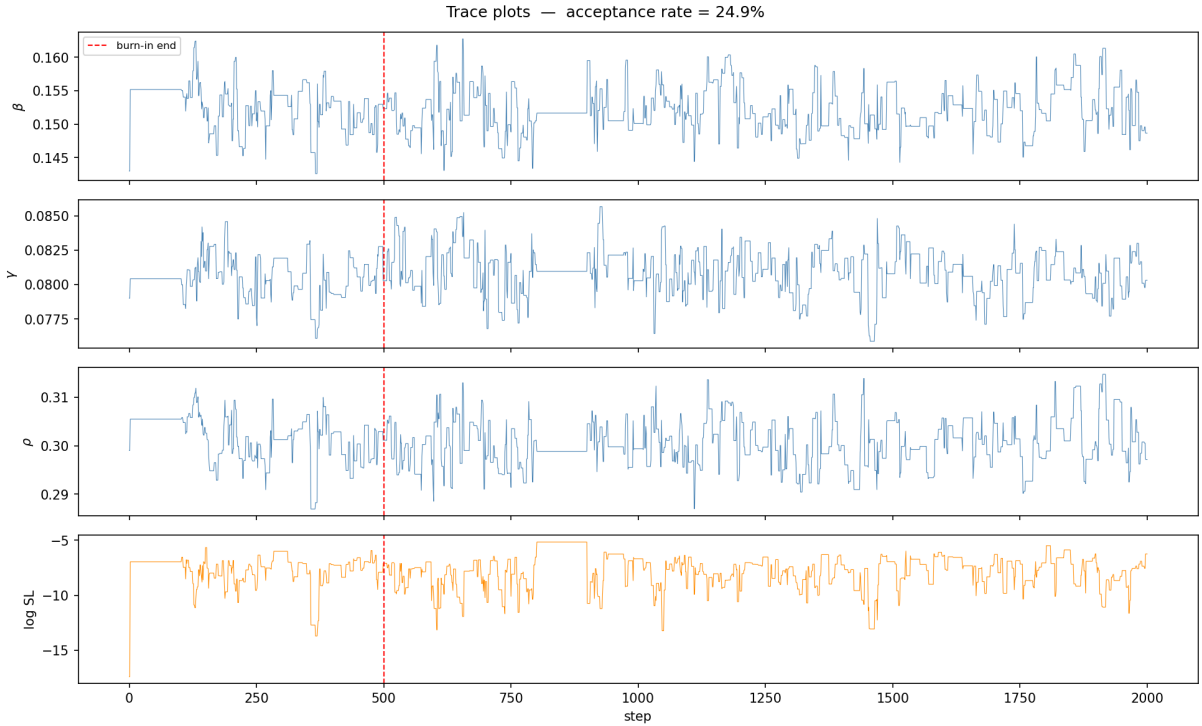


Figure 8: MCMC trace plots. The chain achieves an acceptance rate of 24.0% and mixes across the posterior, though the ACF decays slowly (thinning interval 31), resulting in ESS of 7.5–11.9.

Table 5: Synthetic likelihood MCMC posterior (1500 post-burn-in samples, thinning interval 31, 49 thinned draws; acceptance rate 24.0%).

Param	Mean	Std	5%	Median	95%	ESS
β	0.153	0.003	0.148	0.153	0.158	7.5
γ	0.081	0.002	0.078	0.081	0.084	11.9
ρ	0.301	0.005	0.295	0.300	0.310	10.2

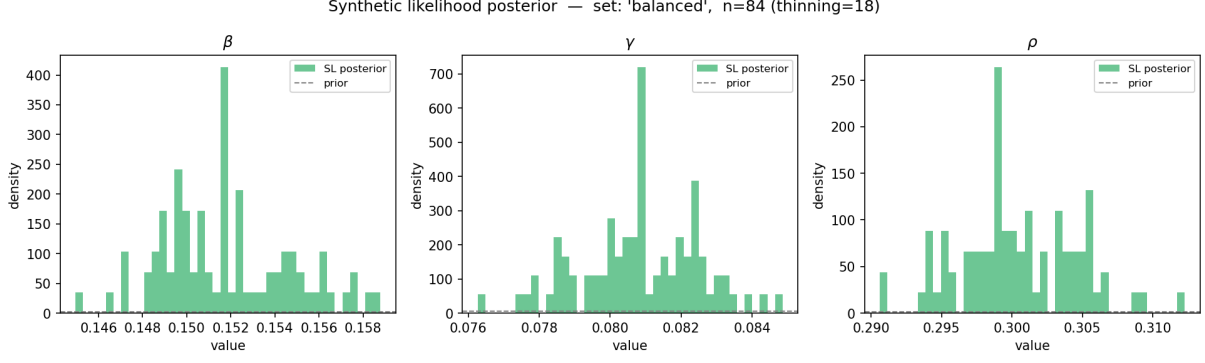


Figure 9: Synthetic likelihood MCMC posterior (green) vs. prior (dashed).

5.5 ABC-MCMC (Marjoram et al., 2003)

At each step, we accept a proposal $\theta^* = \theta_t + \eta$, $\eta \sim \mathcal{N}(0, \Sigma)$ if $\rho(S(x^*), S(x_{\text{obs}})) \leq \varepsilon$, with $R = 40$ replicates per proposal and ε imported from rejection ABC at the 1% quantile. We compare MCMC across the three summary variants: **balanced**, **LASSO-selected**, and **semi-automatic** (with ε recalibrated in the projected space).

We tune the proposal Σ in three stages — original, adaptive $1\times$, and adaptive $0.5\times$ (Roberts and Rosenthal, 2009), with the final selection being the $0.5\times$ as it restores target acceptance rates while achieving $\hat{R} < 1.001$ for all parameters. All results use $N = 30,000$ steps with burn-in $b = 1,000$.

The semi-automatic variant gives the tightest posteriors for β (rel = 0.06) and ρ (rel = 0.08), but at the cost of γ precision (rel = 0.18 vs 0.15–0.16 for balanced and LASSO). This is a consequence of the ridge projection optimising jointly across parameters. As expected, balanced and LASSO perform near-identically. The β – ρ correlation persists across all variants. Fixed ε MCMC cannot resolve this confounding.

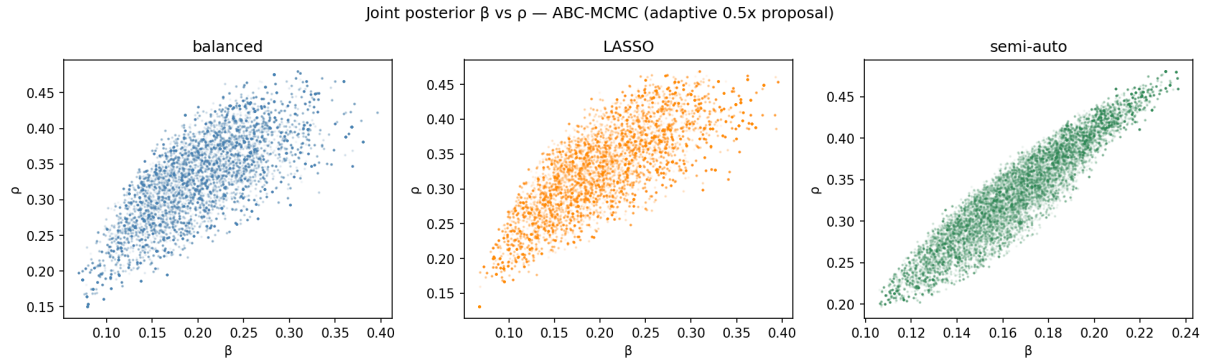


Figure 10: Joint posterior β vs ρ for ABC-MCMC (adaptive $0.5\times$ proposal) across all three summary variants. The positive correlation persists across all variants, confirming β – ρ confounding is an identifiability limitation intrinsic to the aggregate data rather than a sampler artefact.

Table 6: ABC-MCMC posterior shrinkage (adaptive $0.5\times$ proposal, $N = 30,000$ steps, burn-in $b = 1,000$) and split-chain Gelman–Rubin $\hat{R}$.

Variant	β	γ	ρ	$\hat{R}_\beta$	$\hat{R}_\gamma$	$\hat{R}_\rho$
Balanced	0.17	0.15	0.11	1.0003	1.0003	1.0002
LASSO	0.18	0.16	0.12	1.0011	1.0001	1.0000
Semi-auto	0.06	0.18	0.08	1.0000	1.0002	1.0000

5.6 SMC-ABC (Sisson et al., 2007)

ABC-MCMC’s fixed ε limits posterior accuracy and its serially correlated samples reduce ESS. We refine $N = 100$ particles through a decreasing tolerance sequence $\varepsilon_t = Q_{0.75}(\{\rho_i^{(t)}\})$ (Beaumont et al., 2009), producing asymptotically independent samples and perturb particles via $\mathcal{N}(0, \Sigma)$ from the ABC-MCMC, run for $T = 15$ generations. A refined run with Σ halved consistently outperforms the original.

SMC substantially improves on ABC-MCMC across all variants: β shrinkage improves from 0.22 to 0.06 and ρ from 0.28 to 0.08 for the semi-automatic variant, and the β – ρ correlation is substantially reduced. The γ penalty from the ridge projection is also smaller than under MCMC (0.18 vs 0.72), suggesting progressive ε tightening partially recovers lost γ information.

Table 7: SMC-ABC posterior shrinkage (refined $0.5\times$ run, $T = 15$ generations, $N = 100$ particles).

Variant	β	γ	ρ
Balanced	0.17	0.15	0.11
LASSO	0.18	0.16	0.12
Semi-auto	0.06	0.18	0.08

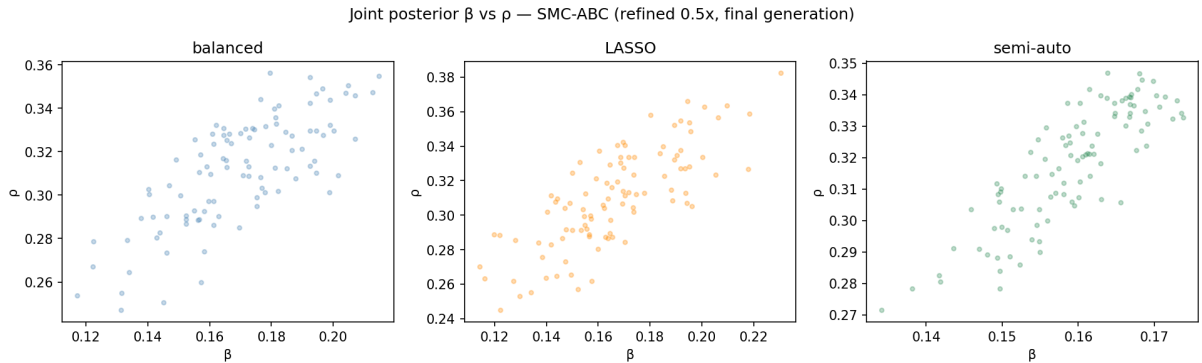


Figure 11: Joint posterior β vs ρ for SMC-ABC (refined $0.5\times$ run, final generation) across all three summary variants. The β – ρ correlation visible under ABC-MCMC is substantially reduced, with particles concentrating onto a narrow region of the joint space.

6 Comparison

Table 8 summarises all methods. Point estimates are consistent across advanced methods ($\hat{\beta} \approx 0.14$ – 0.15 , $\hat{\gamma} \approx 0.079$ – 0.081 , $\hat{\rho} \approx 0.299$ – 0.301). Regression adjustment gives the most reliable uncertainty quantification: large accepted set ($n = 500$), tolerance-robust, and verifiable via sensitivity analysis. The SL posterior is the sharpest, but comes with important caveats: the low ESS (7.5–11.9) means the posterior is estimated from fewer than 20 effectively independent

draws despite 1500 post-burn-in samples, so the quoted standard deviations are unreliable. The tightness likely reflects the high $N_{\text{sim}} = 150$ per step combined with a multivariate normal approximation that may be overconfident for non-Gaussian summary statistics.

Table 8: Posterior standard deviations across all methods. Values in parentheses are shrinkage = posterior std / prior std. [‡]SL std is unreliable due to low ESS (7.5–11.9).

Method	σ_β	σ_γ	σ_ρ	Notes
Prior	0.130	0.052	0.231	–
Rejection ABC				
Balanced ($q = 1\%$)	0.062	0.020	0.061	$n = 100$
LASSO ($q = 1\%$)	0.062	0.020	0.061	All 15 features kept
Semi-auto ($q = 1\%$)	0.023	0.017	0.051	3-d projection
Regression adjustment				
Balanced ($q = 5\%$)	0.024	0.007	0.008	$n = 500$, robust
Synthetic likelihood[‡]				
Balanced	0.003	0.002	0.005	$N_{\text{sim}} = 150$, accept. 24%
ABC-MCMC (adaptive $0.5\times$, $N = 30,000$)				
Balanced	0.0626 (0.48)	0.0204 (0.39)	0.0665 (0.29)	$\hat{R} < 1.001$
LASSO	0.0700 (0.54)	0.0206 (0.40)	0.0691 (0.30)	$\hat{R} < 1.001$
Semi-auto	0.0281 (0.22)	0.0376 (0.72)	0.0637 (0.28)	$\hat{R} < 1.001$
SMC-ABC (refined $0.5\times$, $T = 15$, $N = 100$)				
Balanced	0.0216 (0.17)	0.0078 (0.15)	0.0250 (0.11)	
LASSO	0.0238 (0.18)	0.0086 (0.16)	0.0286 (0.12)	
Semi-auto	0.0082 (0.06)	0.0091 (0.18)	0.0187 (0.08)	

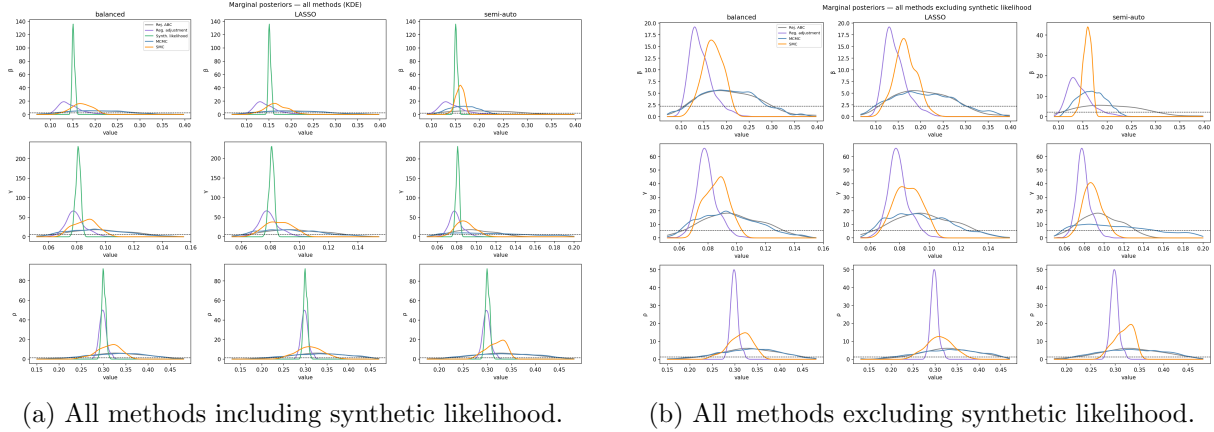


Figure 12: Marginal posteriors (KDE) across all methods. Synthetic likelihood is shown separately as its extreme tightness compresses the scale for other methods.

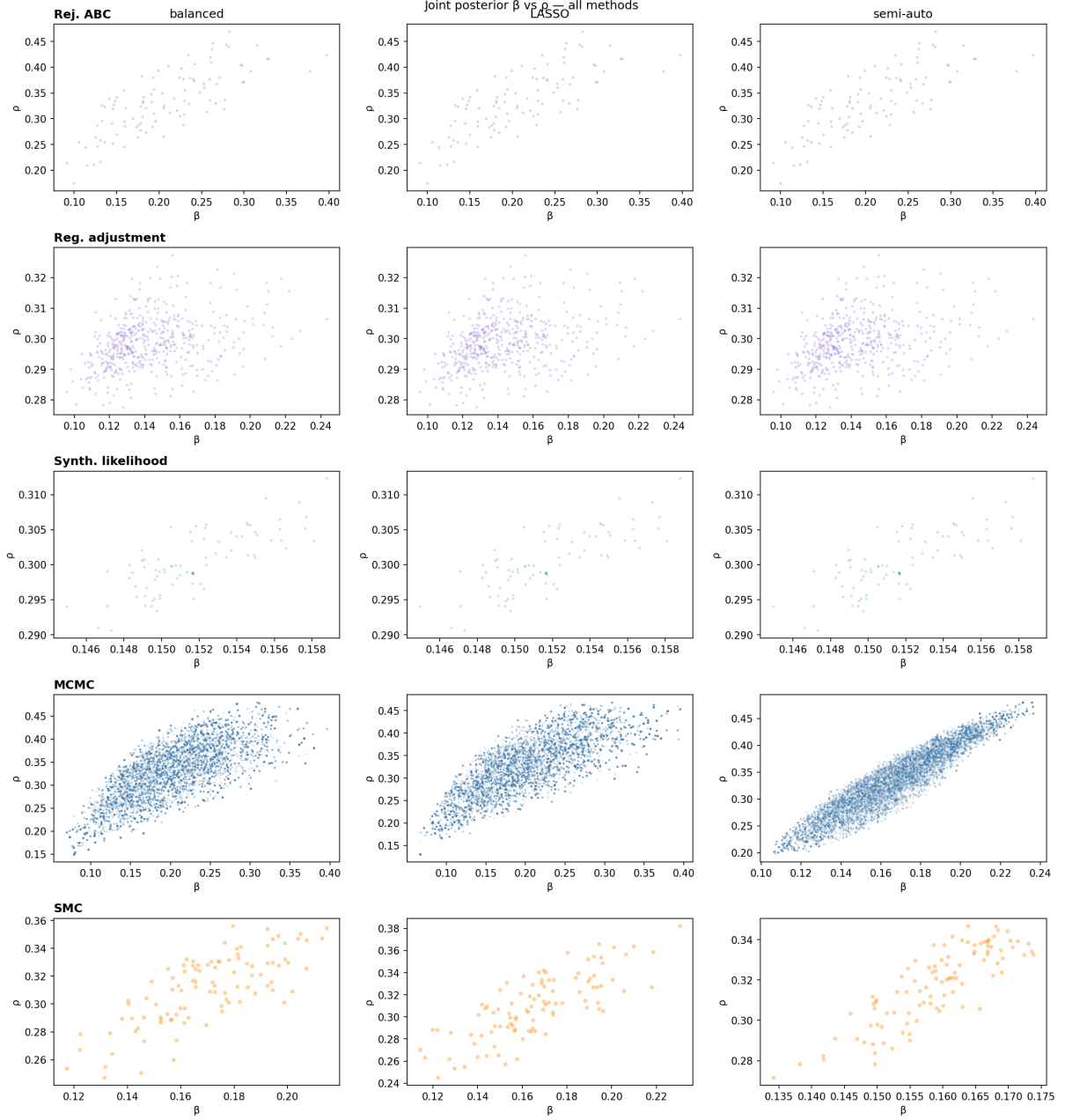


Figure 13: Joint posterior β vs ρ across all methods (rows) and summary variants (columns). Rejection ABC shows a broad positive correlation; regression adjustment and synthetic likelihood concentrate onto a narrow region but may be overconfident; ABC-MCMC preserves the correlation under fixed ε ; SMC-ABC substantially reduces confounding through iterative ε tightening, with the semi-automatic variant producing the tightest joint posterior.

7 Conclusion

[TODO: Summarise: (1) identifiability findings; (2) how each method improves on basic rejection ABC; (3) SL mixing limitations and future work. Suggested length: ~0.5 pages.]

References

- Mark A. Beaumont, Wenyang Zhang, and David J. Balding. Approximate Bayesian computation in population genetics. *Genetics*, 162(4):2025–2035, 2002. doi: 10.1093/genetics/162.4.2025.
- Mark A. Beaumont, Jean-Marie Cornuet, Jean-Michel Marin, and Christian P. Robert. Adaptive approximate Bayesian computation. *Biometrika*, 96(4):983–990, 2009. doi: 10.1093/biomet/asp052.
- Michael G. B. Blum, Matthew A. Nunes, Dennis Prangle, and Scott A. Sisson. A comparative review of dimension reduction methods in approximate Bayesian computation. *Statistical Science*, 28(2):189–208, 2013. doi: 10.1214/12-STS406.
- Paul Fearnhead and Dennis Prangle. Constructing summary statistics for approximate Bayesian computation: Semi-automatic approximate Bayesian computation. *Journal of the Royal Statistical Society: Series B*, 74(3):419–474, 2012. doi: 10.1111/j.1467-9868.2011.01010.x.
- Thilo Gross, Carlos J. Dommar D’Lima, and Bernd Blasius. Epidemic dynamics on an adaptive network. *Physical Review Letters*, 96(20):208701, 2006. doi: 10.1103/PhysRevLett.96.208701.
- Heikki Haario, Eero Saksman, and Johanna Tamminen. An adaptive Metropolis algorithm. *Bernoulli*, 7(2):223–242, 2001. doi: 10.2307/3318737.
- Olivier Ledit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004. doi: 10.1016/S0047-259X(03)00096-4.
- Paul Marjoram, John Molitor, Vincent Plagnol, and Simon Tavaré. Markov chain Monte Carlo without likelihoods. *Proceedings of the National Academy of Sciences*, 100(26):15324–15328, 2003. doi: 10.1073/pnas.0306899100.
- Jonathan K. Pritchard, Mark T. Seielstad, Anna Perez-Lezaun, and Marcus W. Feldman. Population growth of human Y chromosomes: A study of Y chromosome microsatellites. *Molecular Biology and Evolution*, 16(12):1791–1798, 1999. doi: 10.1093/oxfordjournals.molbev.a026091.
- Gareth O. Roberts and Jeffrey S. Rosenthal. Examples of adaptive MCMC. *Journal of Computational and Graphical Statistics*, 18(2):349–367, 2009. doi: 10.1198/jcgs.2009.06134.
- Gareth O. Roberts, Andrew Gelman, and Walter R. Gilks. Weak convergence and optimal scaling of random walk Metropolis algorithms. *Annals of Applied Probability*, 7(1):110–120, 1997. doi: 10.1214/aoap/1034625254.
- Scott A. Sisson, Yanan Fan, and Mark M. Tanaka. Sequential Monte Carlo without likelihoods. *Proceedings of the National Academy of Sciences*, 104(6):1760–1765, 2007. doi: 10.1073/pnas.0607208104.
- Simon N. Wood. Statistical inference for noisy nonlinear ecological dynamic systems. *Nature*, 466:1102–1104, 2010. doi: 10.1038/nature09319.